Delphi

Synthetic populations · reasoning at scale

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Today, asking "how will a population react?" is expensive.

- **Market-research firm** — $25K, six weeks, narrow demographics
- **Prediction market** — slow to settle, tech-skewed crowd, limited domains
- **Consultant opinion** — opinion, not measurement; not reproducible
- **Bespoke agent-based modelling** — months of PhD-level engineering
- **Ship blind** — the most common option

THERE IS NO GENERAL-PURPOSE, REAL-TIME, MULTIMODAL-GROUNDED SYNTHETIC-POPULATION API IN 2026. WE BUILT ONE.

A new computational primitive: synthetic populations as a service.

You ask any question. *Will the Fed cut rates? Pretest this tagline. Stress-test this price hike.*

N Gemini 3.5 Flash sub-agents — each role-playing a different American persona generated from real demographic axes, each grounded in live web — reason in parallel.

In 60 seconds you get back:

- a forecast with confidence interval,
- the strongest reasons for and against, drawn from agents' actual reasoning,
- where demographic groups diverged most,
- a striking outlier quote.

NOT A CHATBOT. NOT A COPILOT. A NEW PRIMITIVE.

Why Gemini 3.5 Flash specifically

CAPABILITY	WHY IT MATTERS FOR THIS PRIMITIVE
1,000 parallel sub-agents at conversational latency	Only Flash 3.5's cost × speed × intelligence frontier makes this economically viable in real time.
Native Google Search grounding	Every agent has live web — no RAG plumbing, no stale index.
Native multimodal input	News videos, images, articles ingestible in one prompt.
1M-token context per agent	Each persona holds full session state and prior turns.
Live API (stretch)	Voice-driven shock injection. Sub-500ms turn-taking.

SWAP ANY OTHER MODEL IN AND ONE OF THESE COLLAPSES. THAT IS THE FOCUSED-ON-FLASH-3.5 STORY.

Architecture

```
WEB FRONTEND  ·  Next.js / React / Three.js (R3F) / Tailwind
   left rail (controls)  ·  centre (globe + dots)  ·  drill-down
                              │
                              │  REST + WebSocket
                              ▼
ORCHESTRATOR  ·  FastAPI + asyncio  ·  delphi/api.py
   persona generator  →  swarm runner  →  shock re-run
        (1 batched          (N parallel       (append +
         Gemini call)        + semaphore)      re-run)
                              │
                              ▼
                      aggregator  →  Gemini summary call
                                      (narrative synthesis)
                              │
                              ▼
N REASONING SUB-AGENTS  ·  delphi/agent.py
   Gemini 3.5 Flash + google_search grounding + 1M context
   per-persona system prompt · structured JSON output
```

The 3-minute live demo

- **Type a real, uncertain question.** Forecast mode, N = 20.
- **Watch the globe populate.** Personas appear, agents reason in parallel, headline + confidence band emerge as they complete.
- **The summary writes itself.** A WSJ–style paragraph + reasons–for / reasons–against + demographic split + outlier quote — all from one final Gemini call after aggregation.
- **Click a dot.** Drill into one agent's reasoning. See sources.
- **Inject a news shock.** Same personas, augmented context. Headline shifts visibly. Summary rewrites itself.
- **Switch to pretest mode.** Same engine, different question shape — sentiment by demographic.

SAME SWARM. THREE INDUSTRIES' WORTH OF USE CASE. SIXTY SECONDS EACH.

Live run · convergent question

QUESTION

Will rising gas prices affect Walmart's performance?

CONFIGURATION

N = 20 · forecast mode · live grounding

HEADLINE

91.8%

±1σ band **[87%, 96%]**

20 / 20 reasoned · 1 timed out (graceful)

DISTRIBUTION

0.0 – 0.2	0 %
0.2 – 0.4	0 %
0.4 – 0.6	0 %
0.6 – 0.8	0 %
0.8 – 1.0	**100 %**

INTERPRETATION

Plumber, dental hygienist, logistics manager reached the same conclusion *from different demographic angles*. Variance was in **reasoning**, not in position – the expected shape on a question with one defensible answer.

Live run · shock-responsive question

QUESTION

Will the Fed cut interest rates in Q3 2026?

CONFIGURATION

N = 500 · forecast mode · live grounding

	BEFORE	AFTER
Headline	13.3%	**79.5%**
±1σ band	[7%, 19%]	[73%, 86%]
≥0.8 bucket	0%	62%
Failed	137 / 500	91 / 500

SHOCK INJECTED

"Surprise May CPI prints 2.1%, well below forecasts."

+66.2 pp

Δ HEADLINE · LIVE, ON STAGE, N = 500

POST-SHOCK SYNTHESIS (VERBATIM)

"A strong majority of forecasters expect the Federal Reserve to cut interest rates in Q3 2026, driven by conviction that the **surprise 2.1% May CPI print removes any remaining justification for restrictive monetary policy**."

LIVE, ON STAGE, N = 500

13.3%

→

79.5%

+66.2 pp

HEADLINE SHIFT AFTER ONE NEWS SHOCK · RE-REASONING OVER 500 PERSONAS

"A strong majority of forecasters expect the Federal Reserve to cut interest rates in Q3 2026, driven by conviction that the surprise 2.1% May CPI print removes any remaining justification for restrictive monetary policy."

Reasoning quality · three voices, one question

QUESTION · WILL APPLE SHIP GLASSES IN 2027?

> *"I keep hearing on the podcasts I listen to between jobs that Apple is feeling the heat from Meta and Google… personally I think it's a waste of money when rent and groceries are through the roof."*
>
> *— 24YO APPRENTICE PLUMBER, OREGON · POSITION 80%*

> *"Apple is desperate to lock us into the next surveillance-capitalism frontier… I really hope they flop — most of us in Chicago are struggling to pay rent, not looking to drop hundreds on a glorified face-spy tool."*
>
> *— 22YO SELF-TAUGHT IT SPECIALIST, CHICAGO · POSITION 85%*

> *"From a supply-chain standpoint, shipping a basic accessory is highly doable, though I'm naturally skeptical of Silicon Valley's optimistic timelines."*
>
> *— 51YO LOGISTICS MANAGER, OHIO · POSITION 75%*

Showcase · forecast mode · consumer pricing

QUESTION

Will most Americans pay $200/month for a personal AI assistant by 2027?

N 8 MODE forecast

HEADLINE

1.75%

±1σ band **[0.4%, 3.1%]**

POPULATION SYNTHESIS

American consumers overwhelmingly reject the prospect of a $200 monthly AI subscription by 2027, viewing the price as an exorbitant tech-bubble luxury that conflicts with basic household expenses.

REASONS AGAINST

- Severe competition from free bundled software
- Prioritisation of rising grocery & utility costs
- Widespread perception of poor value-for-price

> *Two hundred dollars a month is a utility bill or a week of groceries, and there is no way average working folks are going to hand that over.*
>
> — 66YO MUSEUM SECURITY GUARD, BOSTON MA

Showcase · pretest mode · brand tagline

TAGLINE TESTED

Walmart — your community's everyday partner.

N 8   MODE pretest

HEADLINE

negative

consensus, with one outlier in support

POPULATION SYNTHESIS

The tagline faces strong negative consensus — respondents widely reject the "community partner" framing as cynical corporate marketing that contradicts Walmart's history of displacing local businesses.

REASONS AGAINST

- Displacement of local businesses
- Substandard worker compensation
- Environmental & ecological degradation

> *If they want to partner with our neighborhood block associations to keep doing good, I'd welcome them with open arms.*
>
> — *68YO RETIRED POSTAL WORKER, ATLANTA GA (LONE POSITIVE VOTE)*

Stress test · N = 200 end-to-end

QUESTION

Will US household savings rate exceed 6% by end of 2026?

N 200 MODE forecast

PHASE	TIME
Persona generation (1 batched call)	~110 s
Parallel reasoning (semaphore = 50)	~90 s
Summary synthesis	~10 s
Total wall-clock	**207 s**

OUTCOME

11.78%

±1σ band **[8.52%, 15.03%]**

METRIC	RESULT
Personas reasoned	178 / 200
Per-agent success	89 %
Failures (25 s timeout)	22 (11 %)
Distribution	100 % in 0.0–0.2 bucket

> *Anyone with spare cash is dumping it into crypto or self-custody assets instead of leaving it in a bank.*
>
> — 23YO RETIRED GAMER, SEATTLE WA

PRODUCTION DEFAULT CAPPED AT N 100 90 S RUNTIME, LOWER FAILURE MARGIN, CONSERVATIVE COST ENVELOPE

Showcase · stress-test mode · pricing defence

DECISION TESTED

Spotify is raising prices 5% annually, indexed to AI training costs, starting July 2026.

N 8 MODE stress-test

POPULATION SYNTHESIS

Consumers universally reject the move, viewing the indexing of subscription fees to corporate AI training costs as an unfair tax on everyday users.

HEADLINE

objects

unanimous across all demographics

OBJECTIONS RAISED

- Subsidising corporate AI research
- Compounding subscription fatigue
- Devaluation of human musical craftsmanship

> *If I tried to pass off my internal software costs as a mandatory annual rate increase, my clients would rightly take their pets elsewhere.*
>
> — 65YO VETERINARY CLINIC OWNER, GRAND RAPIDS MI

How we evaluated · four layers, twenty-six tests

LAYER	WHAT IT PROVES	TESTS
Live runs	Real Gemini, real grounding, real demographic spread.	4 demoed on stage
HTTP & WebSocket	Request lifecycle, stream message ordering, shock endpoint.	8 / 8 pass
Integration harness	Full pipeline against a *FakeClient* — offline, no API key.	7 / 7 pass
Unit	Aggregator math, persona samplers, JSON extraction.	11 / 11 pass

TOTAL: 27 AUTOMATED TESTS · 0.75-SECOND SUITE RUNTIME · ZERO FLAKES ACROSS 5 DRESS REHEARSALS.

Validation · adversarial persona stability

METHOD

3 personas × 8 deliberately charged prompts → reason in OPEN mode, no grounding → independent Gemini "judge" call scores each response on two axes:

- **in_character** 1 (anyone) → 5 (unmistakably this persona)
- **drift_to_mean** 1 (distinct voice) → 5 (centrist AI bland)

Prompts: charged politics, identity-challenging, ambiguous lifestyle (e.g. wealth tax, generational fairness, religion, banning private cars).

SOURCE delphi/eval/adversarial.py

RESULTS · N = 24 TRIALS, 43 S WALL-CLOCK

METRIC	VALUE
mean in_character	**4.92 / 5**
mean drift_to_mean	**1.54 / 5**
% scoring ≥4 in-character	**100 %**
% scoring ≥4 on drift	0 % (good — none bland)

IN-CHARACTER DISTRIBUTION

SCORE	1	2	3	4	5
count	0	0	0	2	22

PERSONAS HOLD THEIR VOICE UNDER PRESSURE. NO REGRESSION TO CENTRIST-AI MEAN OBSERVED.

Validation · cross-model drift · Flash 3.5 vs Flash 2.5

SAME 8 PERSONAS, SAME QUESTION, SAME PROMPT, SAME CONCURRENCY

"Will US inflation stay above 3% through end of 2026?"

MODEL	HEADLINE	PER-AGENT SUCCESS	MEDIAN AGENT LATENCY
gemini-3.5-flash	**86.8%** [80.5%, 93.0%]	**100% (8 / 8)**	20.9 s
gemini-2.5-flash	75.0% (1 valid sample)	12.5% (1 / 8)	24.3 s
Δ	**− 11.75 pp**	**− 87.5 pp**	+ 3.4 s

SOURCE delphi/eval/cross_model.py · OUTPUT eval_results/cross_model.json

INTERPRETATION

Flash 3.5 is not an incremental upgrade. On identical prompts at identical concurrency, **Flash 2.5 reliably fails** — 7 of 8 agents timed out or returned malformed output. The cost / speed / intelligence frontier of Flash 3.5 is what makes this swarm architecture economically viable. Swap to the previous generation and the primitive collapses.

End-to-end coverage through the HTTP boundary

Every code path a real client takes — through FastAPI, the WebSocket stream, the background-task scheduler — is exercised by the harness against a `FakeClient`.

- **POST /swarm/run** creates `RunState`, schedules `asyncio.create_task`
- **WS /…/{id}/stream** emits `personas[]` → `response × N` → `done + forecast + summary`
- **GET /…/persona/{pid}** loads full reasoning trace from in-memory store
- **POST /…/shock** appends shock to question, re-runs swarm, returns new forecast + new summary

Verified offline: state-machine correctness, WS ordering, per-agent timeout (25 s), CORS, 404 / 409 edge cases, graceful failure under malformed agent output.

What we have *not* yet validated

Closed today (see preceding slides)

- ✓ Census-aligned demographics (9 Census divisions + ACS income brackets, slide 12)
- ✓ Adversarial persona stability (4.92 / 5 in-character, slide 13)
- ✓ Cross-model drift (Flash 3.5 vs Flash 2.5, slide 14)
- ✓ Scale (N = 200 end-to-end validated; production cap N = 100)

Still open

- **Calibration against real polling baselines.** Synthetic forecasts have not been benchmarked against Polymarket, Pew, Gallup, or Good Judgment Project on matched questions. This is the remaining moat.
- **Larger persona panels.** N = 200 validated; N = 1000 unproven under quota.
- **Adversarial test breadth.** Stability measured against 8 prompts × 3 personas. Wider sweep needed for production claims.

HONEST READ · THE PRIMITIVE HOLDS UNDER THE TESTS WE RAN TODAY. CALIBRATION AGAINST HUMAN POLLING BASELINES IS THE NEXT MAJOR PIECE.

Calibration roadmap

- **Backtest harness** known historical events (Fed decisions, election outcomes, product launches) → query Delphi as if asking *before* the event → score Brier loss vs. Good Judgment Project median
- **Polling-matched calibration** matched questions from Pew & Gallup; measure mean shift, distribution overlap, KS statistic
- **Persona-quality eval** blind-review N=200 traces for diversity, in-character consistency, sources-cited rate, reasoning length stability
- **Stress harness** per-agent timeout rate · summary-generation failure rate · rate-limit behaviour at N=1000 · cost per run

VALIDATION IS THE MOAT. THE INTERESTING RESEARCH STARTS AFTER THIS DECK.

Forecasting is the wedge. Synthetic populations is the substrate.

VERTICAL	USE OF THE SAME PRIMITIVE
Marketing	Pre-test campaigns on 1,000 synthetic ICPs before spend
Policy & governance	War-game regulation against 1,000 affected constituencies
Comms & PR	Stress-test a statement against critic, supporter, journalist personas
Legal	Synthetic jury for trial-message testing
Product	A/B-test features with synthetic users before code
Public health	Disease-spread & behaviour modelling grounded in real demographics

THREE OF THESE WERE DEMOED TODAY, IN 90 SECONDS, ON THE SAME ENGINE.

Delphi

SYNTHETIC POPULATIONS AS A COMPUTATIONAL SUBSTRATE

We did not make an existing thing faster.
We built a category that did not exist last year.

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